Physics Department
Fisika Departement

University of Pretoria

Semester Test 1 **FSK176** *Semestertoets 1*

18 August/Augustus 2010

Surname and Initials:___________________________ Student Number: _____________________
Van en voorletters: *Studentenommer:*

Signature: ___________________________
Handtekening:

Time : 2 hours Total : 70 Max
Tyd: 2 ure *Totaal: 70 Maks*

Answer all questions.	*Beantwoord alle vrae.*
Programmable calculators may NOT be used.	*Programeerbare sakrekenaars MAG NIE gebruik word nie.*
No textbooks allowed.	*Geen handboeke word toegelaat.*
A formula sheet is given below	*'n Formuleblad verskyn hieronder.*
All test and examination regulations of the University of Pretoria are applicable	*Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

Formulas / Formule

Circle: Circumference $=2\pi r$ = omtrek

Area = πr^2

Sphere: Area = $4\pi r^2$

Volume = $^4/_3\pi r^3$

Cylinder / Silinder

Area = $2(\pi r^2) + 2\pi rh$

Volume = $\pi r^2 h$

Trapesium: Area = $½(a+b)h$

$\vec{a}\cdot\vec{b} = ab\cos\phi$

$\vec{a}\cdot\vec{b} = a_x b_x + a_y b_y + a_z b_z$

$$\vec{a}\times\vec{b} = \left(a_y b_z - b_y a_z\right)\hat{i} + \left(a_z b_x - b_x a_z\right)\hat{j} + \left(a_x b_y - b_y a_x\right)\hat{k}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$v_{avg} = ½\left(v_0 + v\right)$

$v = v_0 + at$

$x = x_0 + v_0 t + ½at^2$

$x = x_0 + v_{avg} t$

$v^2 = v_o^2 + 2a\left(x - x_0\right)$

$x - x_0 = ½\left(v_0 + v\right)t$

$x - x_0 = vt - ½at^2$

1 ml = 1 cm^3

✂--✂---✂---

SEMESTER TEST 1 **FSK 176** **SEMESTERTOETS 1**

18 August/Augustus 2010

Surname and Initials:___________________________ Student Number: _____________________
Van en voorletters: *Studentenommer:*

Signature: ___________________________ Venue:_____________________
Handtekening: *Lokaal:*

Question 1 **[4+2+3]**

Water is poured into a container that has a leak. The mass m of the water is given as a function of time t by $m = 5.00t^{0.8} - 3.00t + 20.00$, with $t \geq 0$, m in grams, and t in seconds.

Water word in 'n houer gegooi wat lek. Die massa, m, van die water word as 'n funksie van tyd t gegee, $m = 5.00t^{0.8} - 3.00t + 20.00$*, waar* $t \geq 0$*, m is in gram, en t is in sekondes.*

(a) At what time is the water mass greatest?
Teen watter tyd is die water massa die grootste?

(b) What is that greatest mass?
Wat is die grootste massa?

(c) In kilograms per minute, what is the rate of mass change at $t = 5.00$ s?
In kilogram per minuut, wat is die tempo van verandering van massa by $t = 5.00s$*?*

Question 2 **[4+3]**

One cubic centimeter of a typical cumulus cloud contains 50 to 500 water drops, which have a typical radius of $10\ \mu m$.

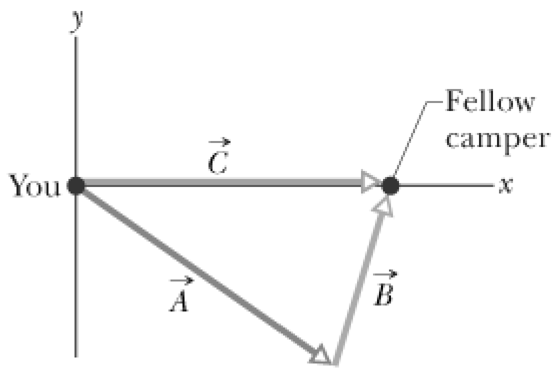
Een kubieke centimeter van 'n tipies kumulus wolk het tussen 50 en 500 waterdruppels, elk met 'n tipiese radius van 10 μm.

(a) What is the maximum volume of water, in cubic meters, in a cylindrical cumulus cloud of height 3.0 km and radius 1.0 km?
Wat is die maksimum volume water, in kubieke meter, in 'n silindriese kumulus wolk met 'n hoogte van 3.0 km en radius van 1.0 km?

(b) How many litres of water is this? Hint: $1\ ml = 1\ cm^3$.
Hoeveel liter water is dit? Wenk: 1 ml = 1 cm^3.

Question 3 **[3+3]**

A fellow camper is to walk away from you in a straight line (vector $\vec{A}$), turn, walk in a second straight line (vector $\vec{B}$) and then stop. How far must you walk in a straight line (vector $\vec{C}$) to reach her? The three vectors (shown in the Figure) are related by $\vec{C} = \vec{A} + \vec{B}$. $\vec{A}$ has a magnitude of 22.0 m and is directed at an angle of $-47.0°$ (clockwise) from the positive direction of the x axis. $\vec{B}$ has a magnitude of 17.0 m and is directed counterclockwise from the positive direction of the x axis by angle ϕ. $\vec{C}$ is in the positive direction of the x axis.



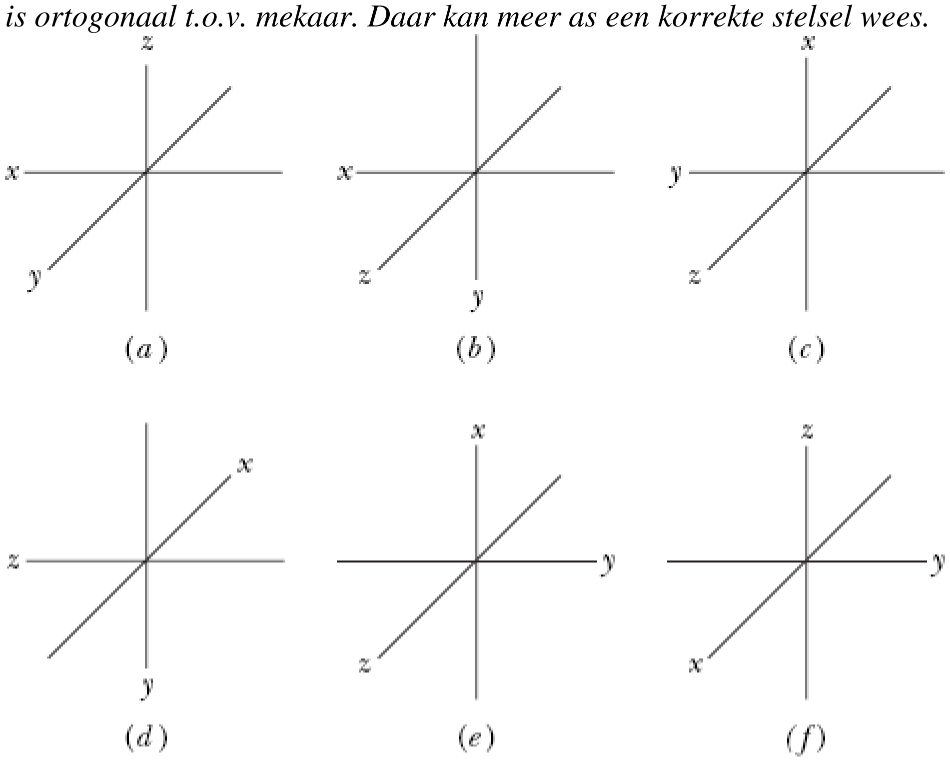
'n Kampeerde loop in 'n reguit-lyn weg van jou af (vektor $\vec{A}$*), draai, en loop weer in 'n reguit-lyn (vektor* $\vec{B}$*) voordat sy stop. Hoe ver moet jy loop in 'n reguit-lyn (vektor* $\vec{C}$*) om by haar uit te kom. Die drie vektore (soos aangedui in die Figuur) het 'n verwantskap* $\vec{C} = \vec{A} + \vec{B}$. $\vec{A}$ *het 'n groote van 22.0 m en is gerig teen 'n hoek van* $-47.0°$ *(kloksgewys) met die positiewe x-as se rigting.* $\vec{B}$ *het 'n groote van 17.0 m en is gerig teen 'n hoek van* ϕ *(antikloksgewys) met die positiewe x-as se rigting.* $\vec{C}$ *is langs die positiewe x-as.*

(a) What is the angle ϕ which $\vec{B}$ makes with the positive x axis?
Wat is die hoek ϕ wat $\vec{B}$ met die x-as maak?

(b) What is the magnitude of $\vec{C}$?
Wat is die groote van $\vec{C}$?

Question 4 **[3]**

Which of the arrangements of axes in the Figure can be labeled "right-handed coordinate system"? As usual, each axis label indicates the positive side of the axis and all the axes are orthogonal to each other. There can be more than one correct system.
Watter van die volgende assestelsels in die figuur hieronder kan as "regterhandse koördinaat stelsel" genoem word. Soos gewoonlik word die positiewe kant van elke as aangedui en al die asse is ortogonaal t.o.v. mekaar. Daar kan meer as een korrekte stelsel wees.



Question 5 **[2+2+3]**

Two vectors are given by $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ and $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$. Determine the following:

Twee vektore word deur $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ *en* $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$ *gegee. Bepaal die volgende:*

(a) $\vec{a} \times \vec{b}$

(b) $\vec{a} \cdot \vec{b}$

(c) $\left(\vec{a} + \vec{b}\right) \cdot \vec{b}$

Question 6 **[6]**

If $\vec{B}$ is added to $\vec{A}$, the result is $6.0\hat{i} + 1.0\hat{j}$ m. If $\vec{B}$ is subtracted from $\vec{A}$, the result is $-4.0\hat{i} + 7.0\hat{j}$ m. What is the magnitude and direction of $\vec{A}$?

Indien $\vec{B}$ *by* $\vec{A}$ *getel word, is die resultaat* $6.0\hat{i} + 1.0\hat{j}$ *m. Indien* $\vec{B}$ *van* $\vec{A}$ *afgetrek word, is die resultaat* $-4.0\hat{i} + 7.0\hat{j}$ *m. Wat is die groote en rigting van* $\vec{A}$*?*

Question 7 **[6]**

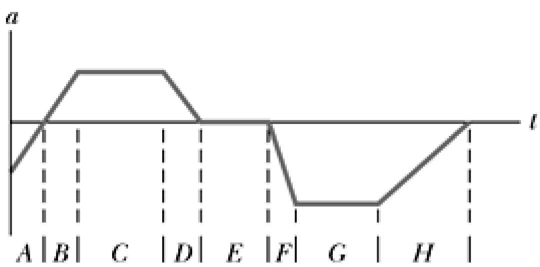
When the acceleration is constant, the average and instantaneous acceleration are equal: $a = a_{avg} = \frac{v - v_0}{t - 0}$. We can also write the average velocity as $v_{avg} = \frac{x - x_0}{t - 0}$. Using these two equations show that $x - x_0 = v_0 t + \frac{1}{2}at^2$.

Wanneer die versnelling konstant is, is die gemiddelde en oombliklike versnelling dieselfde: $a = a_{gem} = \frac{v - v_0}{t - 0}$. *Ons kan ook die gemiddelde snelheid as* $v_{gem} = \frac{x - x_0}{t - 0}$ *skryf. Deur gebruik te maak van hierdie twee vergelykings, bewys dat* $x - x_0 = v_0 t + \frac{1}{2}at^2$.

Question 8 **[3]**

The figure gives the acceleration, $a(t)$, of a Chihuahua as it chases a German Shepherd along the road (x-axis). In which of the time periods indicated does the Chihuahua move at constant speed?

Die figuur gee die versnelling, a(t), van 'n Chihuahua soos hy agter 'n Duitse Herdershond in die straat hardloop (x-as). In watter van die tydsintervalle aangedui is die Chihuahua se spoed konstant?



Question 9 **[3+3+2]**

The position of a particle is given by $x = 20t - 5t^3$, where x is in meters and t is in seconds.
Die posisie van 'n deeltjie word gegee deur $x = 20t - 5t^3$, waar x in meter en t in sekondes is.

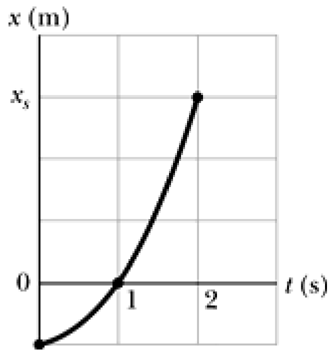
(a) When, if ever, is the particle's velocity zero?
Wanneer, indien ooit, is die deeltjie se snelheid nul?

(b) When is its acceleration a zero?
Wanneer is die versnelling a nul?

(c) For what time range is the acceleration, a, negative?
Vir watter tydsinterval is die versnelling, a, negatief?

Question 10 **[5+2]**

The figure depicts the motion of a particle moving along an x axis with a constant acceleration. The figure's vertical scaling is set by $x_s = 6.0$ m.
Die figuur dui die beweging van 'n deeltjie met konstante versnelling langs 'n x-as aan. Die figuur se vertikale-skaal is gekies sodat $x_s = 6.0$ m.



(a) What is the magnitude of the particle's acceleration?
Wat is die groote van die deeltjie se versnelling?

(b) What is the direction of the particle's acceleration?
Was is die rigting van die deeltjie se versnelling?

Question 11 **[3+2]**

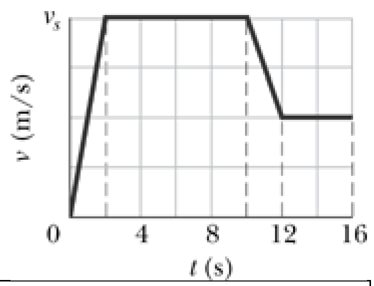
A ball is thrown vertically downward from the top of a 36.6 m-tall building. The ball passes the top of a window that is 12.2 m above the ground 2.00 s after being thrown.
'n Bal word vertikaal afwaarts vanaf 'n 36.6 m hoë gebou gegooi. Die bal gaan 2.00 s later verby die bokant van 'n venster wat 12.2 m bo die grond is.

(a) What is the initial velocity of the ball?
Wat is die aanvanklike snelheid van die bal?

(b) What is the velocity of the ball as it passes the top of the window?
Wat is die snelheid van die bal as dit by die bokant van die venster verby gaan?

Question 12 **[5]**

The figure shows the velocity time graph for a athlete. How far does the athlete travel in 16 s if v_s is 8.0 m/s?
Die figuur wys die snelheid-tyd grafiek vir 'n atleet. Hoe ver het die atleet beweeg in 16 s indien v_s 8.0 m/s *is?*



That's all for now!
Dis al vir nou!

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Area $= 2(\pi r^2) + 2\pi rh$

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Trapesium: Area $= ½(a+b)h$

$\vec{a}\cdot\vec{b} = ab\cos\phi$

$\vec{a}\cdot\vec{b} = a_x b_x + a_y b_y + a_z b_z$

$$\vec{a}\times\vec{b} = (a_y b_z - b_y a_z)\hat{i} + (a_z b_x - b_z a_x)\hat{j} + (a_x b_y - b_x a_y)\hat{k}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$v_{avg} = ½(v_0 + v)$

$v = v_0 + at$

$x = x_0 + v_0 t + ½at^2$

$x = x_0 + v_{avg} t$

$v^2 = v_o^2 + 2a(x - x_0)$

$x - x_0 = ½(v_0 + v)t$

$x - x_0 = vt - ½at^2$

1 ml = 1 cm^3

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Water is poured into a container that has a leak. The mass m of the water is given as a function of time t by $m = 5.00t^{0.8} - 3.00t + 20.00$, with $t \geq 0$, m in grams, and t in seconds.

Water word in 'n houer gegooi wat lek. Die massa, m, van die water word as 'n funksie van tyd t gegee, $m = 5.00t^{0.8} - 3.00t + 20.00$*, waar* $t \geq 0$*, m is in gram, en t is in sekondes.*

(a) At what time is the water mass greatest?
Teen watter tyd is die water massa die grootste?

$$m = 5.00t^{0.8} - 3.00t + 20.00$$
$$\frac{dm}{dt} = 4.00t^{-0.2} - 3.00 = 0 \quad ✓ ✓$$
$$t^{0.2} = \frac{4.00}{3.00} \quad ✓$$
$$t = 4.2 \text{ s} \quad ✓$$

(b) What is that greatest mass?
Wat is die grootste massa?

$$m = 5.00t^{0.8} - 3.00t + 20.00$$
At $t = 4.2$ s ✓
$$m = 5.00(4.2)^{0.8} - 3.00(4.2) + 20.00$$
$$= 23.2 \text{ g} \quad ✓$$

(c) In kilograms per minute, what is the rate of mass change at $t = 5.00$ s?
In kilogram per minuut, wat is die tempo van verandering van massa by t = 5.00s?

$$\frac{dm}{dt} = 4.00t^{-0.2} - 3.00$$
$$= 4.00(5)^{-0.2} - 3.00 \quad ✓$$
$$= -0.10 \text{ g/s} \quad ✓$$
$$= -6.05 \times 10^{-3} \text{ kg/min} \quad ✓$$

Question 2 **[4+3]**

One cubic centimeter of a typical cumulus cloud contains 50 to 500 water drops, which have a typical radius of 10 µm.

Een kubieke centimeter van 'n tipies kumulus wolk het tussen 50 en 500 waterdruppels, elk met 'n tipiese radius van 10 µm.

(a) What is the maximum volume of water, in cubic meters, in a cylindrical cumulus cloud of height 3.0 km and radius 1.0 km?
Wat is die maksimum volume water, in kubieke meter, in 'n silindriese kumulus wolk met 'n hoogte van 3.0 km en radius van 1.0 km?

Volume of the cloud (cylinder) $V_c = h\,\pi r^2 = (3000\text{ m})\pi(1000\text{ m})^2 = 9.4 \times 10^9\text{ m}^3$. $1\text{ m}^3 = 1 \times 10^6\text{ cm}^3$ 1 m^3 contains up to 500×10^6 water drops, Entire cloud contains up to $V_c \times 500 \times 10^6$ drops/m^3 $= 4.7 \times 10^{19}$ drops.	✓ ✓

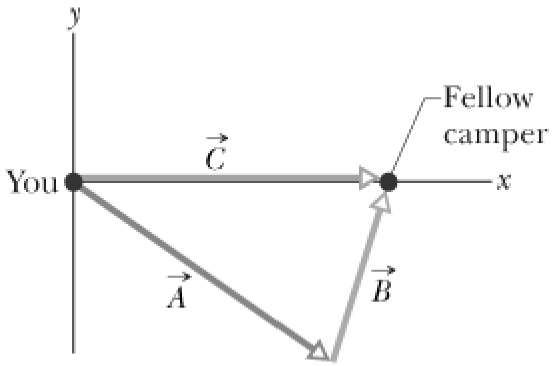
Volume of each drop is $V_{drop} = \frac{4}{3}\pi(10 \times 10^{-6}\text{ m})^3 = 4.2 \times 10^{-15}\text{ m}^3$, then the max volume of water in a cloud is $2\times10^4\ \text{m}^3$.	✓ ✓

(b) How many litres of water is this? Hint: 1 ml = 1 cm^3.
Hoeveel liter water is dit? Wenk: 1 ml = 1 cm^3.

$1\text{ ml} = 1\text{ cm}^3$ $1\text{ L} = 1000\text{ ml} = 1\times10^3\text{cm}^3 = 1\times10^{-3}\text{m}^3$ The amount of water estimated in part (a) would be 2×10^7 L.	✓ ✓ ✓

Question 3 **[3+3]**

A fellow camper is to walk away from you in a straight line (vector $\vec{A}$), turn, walk in a second straight line (vector $\vec{B}$) and then stop. How far must you walk in a straight line (vector $\vec{C}$) to reach her? The three vectors (shown in the Figure) are related by $\vec{C} = \vec{A} + \vec{B}$. $\vec{A}$ has a magnitude of 22.0 m and is directed at an angle of $-47.0°$ (clockwise) from the positive direction of the x axis. $\vec{B}$ has a magnitude of 17.0 m and is directed counterclockwise from the positive direction of the x axis by angle ϕ. $\vec{C}$ is in the positive direction of the x axis.



'n Kampeerde loop in 'n reguit-lyn weg van jou af (vektor $\vec{A}$), draai, en loop weer in 'n reguit-lyn (vektor $\vec{B}$) voordat sy stop. Hoe ver moet jy loop in 'n reguit-lyn (vektor $\vec{C}$) om by haar uit te kom. Die drie vektore (soos aangedui in die Figuur)het die verwantskap $\vec{C} = \vec{A} + \vec{B}$. $\vec{A}$ het 'n groote van 22.0 m en is gerig teen 'n hoek van $-47.0°$ (kloksgewys) met die positiewe x-as se rigting. $\vec{B}$ het 'n groote van 17.0 m en is gerig teen 'n hoek van ϕ (antikloksgewys) met die positiewe x-as se rigting. $\vec{C}$ is langs die positiewe x-as.

(a) What is the angle ϕ which $\vec{B}$ makes with the positive x axis?
Wat is die hoek ϕ wat $\vec{B}$ met die x-as maak?

Since $\vec{C}$ is directed along the x axis, the sum of the y-components must equal 0. $C_y = A_y + B_y = 0$ $= A\sin(-47°) + B\sin\phi$ $-22\sin(-47°) = 17\sin\phi$ $\sin\phi = \dfrac{-22\sin(-47°)}{17}$ $\phi = 71°$	 ✓ ✓ ✓

(b) What is the magnitude of $\vec{C}$?
Wat is die groote van $\vec{C}$?

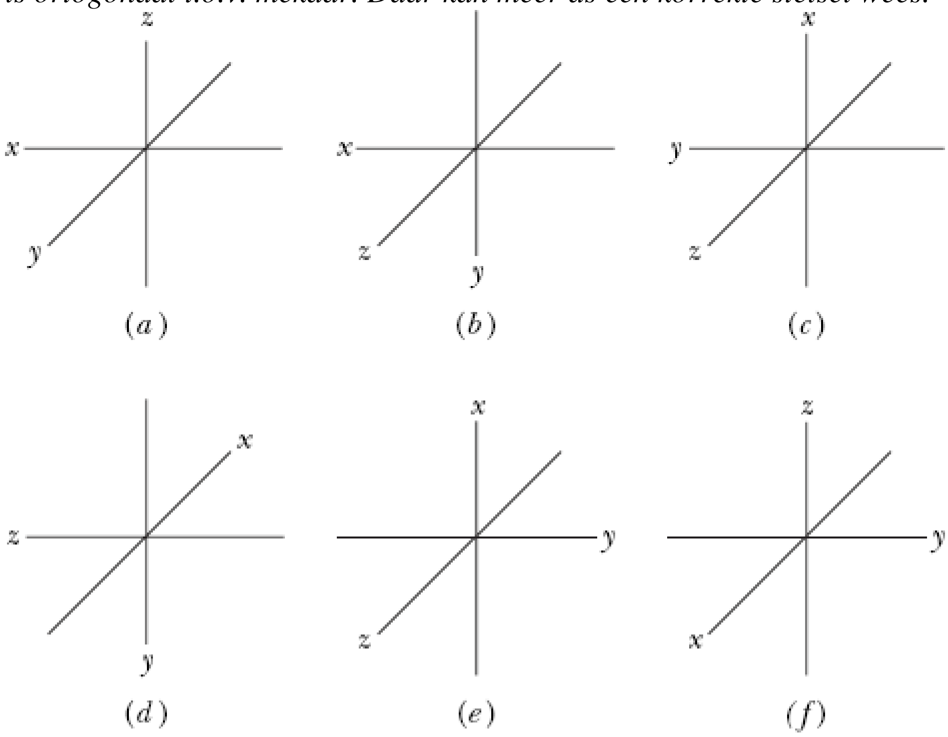
Adding the x-components together we get that	✓

$C_x = A_x + B_x = A\cos(-47°) + B\cos(71°)$	✓
$= 22\cos(-47°) + 17\cos(71°)$	
$= 20.5$ m	✓
Since $\vec{C}$ is directed along the x-axis, the magnitude of $\vec{C}$ is 20.5 m.	

Question 4 **[3]**

Which of the arrangements of axes in the Figure can be labeled "right-handed coordinate system"? As usual, each axis label indicates the positive side of the axis and all the axes are orthogonal to each other. There can be more than one correct system.

Watter van die volgende assestelsels in die figuur hieronder kan as "regterhandse koördinaat stelsel" genoem word. Soos gewoonlik word die positiewe kant van elke as aangedui en al die asse is ortogonaal t.o.v. mekaar. Daar kan meer as een korrekte stelsel wees.



a,b,c,f (-1 ✓ for each incorrect)

Question 5 **[2+2+3]**

Two vectors are given by $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ and $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$. Determine the following:

Twee vektore word deur $\vec{a} = 3.0\hat{i} + 5.0\hat{j}$ en $\vec{b} = 2.0\hat{i} + 4.0\hat{j}$ gegee. Bepaal die volgende:

(a) $\vec{a}\times\vec{b}$

$\vec{a}\times\vec{b} = (a_x b_y - a_y b_x)\ \hat{k}$

$= [(3.0)(4.0) - (5.0)(2.0)]\hat{k} = 2.0\hat{k}$. ✓✓

(b) $\vec{a}\cdot\vec{b}$

$\vec{a}\cdot b = a_x b_x + a_y b_y$

$= (3.0)(2.0) + (5.0)(4.0) = 26.$ ✓✓

(c) $\left(\vec{a}+\vec{b}\right)\cdot\vec{b}$

$\vec{a}+\vec{b} =$ (3.0 + 2.0) $\hat{\mathrm{i}}$ + (5.0 + 4.0) $\hat{\mathrm{j}}$ ✓
Then
$(\vec{a}+\vec{b})\cdot\vec{b}$ = (5.0) (2.0) + (9.0) (4.0) = 46 . ✓✓

Question 6 **[6]**

If $\vec{B}$ is added to $\vec{A}$, the result is $6.0\hat{i}+1.0\hat{j}$ m. If $\vec{B}$ is subtracted from $\vec{A}$, the result is $-4.0\hat{i}+7.0\hat{j}$ m. What is the magnitude and direction of $\vec{A}$?

Indien $\vec{B}$ by $\vec{A}$ getel word, is die resultaat $6.0\hat{i}+1.0\hat{j}$ m. Indien $\vec{B}$ van $\vec{A}$ afgetrek word, is die resultaat $-4.0\hat{i}+7.0\hat{j}$ m. Wat is die groote en rigting van $\vec{A}$?

$\vec{A}+\vec{B} = 6.0\,\hat{\mathrm{i}} + 1.0\,\hat{\mathrm{j}}$ and $\vec{A}-\vec{B} = -4.0\,\hat{\mathrm{i}} + 7.0\,\hat{\mathrm{j}}$.
Adding these two equns
2 $\vec{A}$ $=2.0\,\hat{\mathrm{i}} + 8.0\,\hat{\mathrm{j}}$. Therefore
$\vec{A} = 1.0\,\hat{\mathrm{i}} + 4.0\,\hat{\mathrm{j}}$. ✓✓
The Pythagorean theorem then leads to $A=\sqrt{(1.0)^2+(4.0)^2}=4.1\,\mathrm{m}$. ✓ ✓
Direction above the $+x$ axis: $\phi = \tan^{-1}\frac{4}{1} = 75°$ ✓✓

Question 7 **[6]**

When the acceleration is constant, the average and instantaneous acceleration are equal: $a = a_{avg} = \frac{v-v_0}{t-0}$. We can also write the average velocity as $v_{avg} = \frac{x-x_0}{t-0}$. Using these two equations show that $x - x_0 = v_0 t + ½at^2$.

Wanneer die versnelling konstante is, is die gemiddelde en oombliklike versnelling dieselfde: $a = a_{gem} = \frac{v-v_0}{t-0}$. Ons kan ook die gemiddelde snelheid as $v_{gem} = \frac{x-x_0}{t-0}$ skryf. Deur gebruik te maak van hierdie twee vergelykings, bewys dat $x - x_0 = v_0 t + ½at^2$.

$a = a_{avg} = \frac{v-v_0}{t-0}$ ✓

$v = v_0 + at$

$v_{avg} = \frac{x-x_0}{t-0}$ ✓

$x = x_0 + v_{avg}t$

But $v_{avg} = ½\left(v+v_0\right)$. ✓ Substituting this for v_{avg}:

$x = x_0 + \left[½\left(v+v_0\right)\right]t$ ✓

From the first equation, we substitute for v and rearrange the terms:

$$x = x_0 + \left[\frac{1}{2}(v+v_0)\right]t$$

$$x - x_0 = \frac{1}{2}vt + \frac{1}{2}v_0 t$$

$$= \frac{1}{2}(v_0 + at)t + \frac{1}{2}v_0 t \quad ✓$$

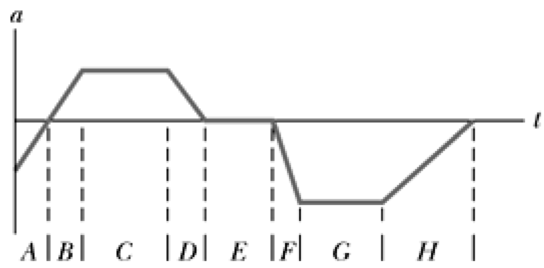
$$= v_0 t + \frac{1}{2}at^2$$

✓

Question 8 **[3]**

The figure gives the acceleration, $a(t)$, of a Chihuahua as it chases a German Shepherd along the road (x-axis). In which of the time periods indicated does the Chihuahua move at constant speed?

Die figuur gee die versnelling, a(t), van 'n Chihuahua soos hy agter 'n Duitse Herdershond in die straat hardloop (x-as). In watter van die tydsintervalle aangedui is die Chihuahua se spoed konstant?



E✓✓✓

Question 9 **[3+3+2]**

The position of a particle is given by $x = 20t - 5t^3$, where x is in meters and t is in seconds.

Die posisie van 'n deeltjie word gegee deur $x = 20t - 5t^3$, waar x in meter en t in sekondes is.

(a) When, if ever, is the particle's velocity zero?
Wanneer, indien ooit, is die deeltjie se snelheid nul?

$$x = 20t - 5t^3$$

$$v = \frac{dx}{dt} = 20 - 15t^2 = 0 \quad ✓✓$$

$$t = \sqrt{\frac{20}{15}} = \pm 1.16 \text{ s} \quad ✓$$

$$= \pm 1.2 \text{ s}$$

(b) When is its acceleration a zero?
Wanneer is die versnelling a nul?

$$v = 20 - 15t^2$$

$$a = \frac{dv}{dt} = -30t = 0 \quad ✓✓$$

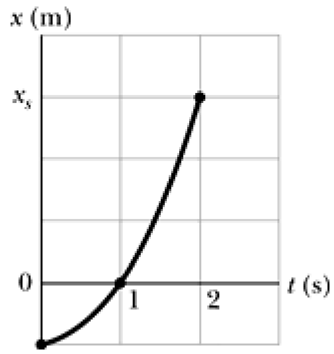
$$t = 0 \text{ s} \quad ✓$$

(c) For what time range is the acceleration, a, negative?
Vir watter tydsinterval is die versnelling, a, negatief?

For all $t > 0$ s ✓✓

Question 10 **[5+2]**

The figure depicts the motion of a particle moving along an x axis with a constant acceleration. The figure's vertical scaling is set by $x_s = 6.0$ m.
Die figuur dui die beweging van 'n deeltjie met konstante versnelling langs 'n x-as aan. Die figuur se vertikale-skaal is gekies sodat $x_s = 6.0$ m.



(c) What is the magnitude of the particle's acceleration?
Wat is die groote van die deeltjie se versnelling?

At t = 0 s, x_0 = -2.0 m At t = 1 s, x = 0 m At t = 2 s, x = 6 m $x - x_0 = v_0 t + ½at^2$ At t = 1s $0-(-2) = v_0(1) + ½a(1)^2$ -1 At t = 2 s $6-(-2) = v_0(2) + ½a(2)^2$ -2 Solving simultaneously: Equn 1 x 2 – equn 2: $-4 = -2a$ $a = 4$ m/s^2	✓ ✓ ✓ ✓ ✓

(d) What is the direction of the particle's acceleration?
Was is die rigting van die deeltjie se versnelling?

in the positive x direction ✓✓

Question 11 **[3+2]**

A ball is thrown vertically downward from the top of a 36.6 m-tall building. The ball passes the top of a window that is 12.2 m above the ground 2.00 s after being thrown.
'n Bal word vertikaal afwaarts vanaf 'n 36.6 m hoë gebou gegooi. Die bal gaan 2.00 s later verby die bokant van 'n venster wat 12.2 m bo die grond is.

(a) What is the initial velocity of the ball?
Wat is die aanvanklike snelheid van die bal?

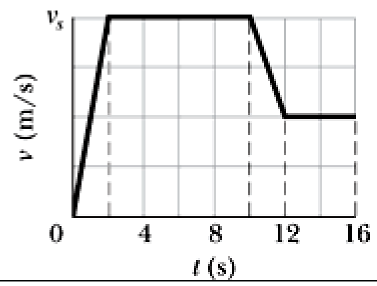
y_0 = 36.3 m y = 12.2 m a= -9.8 m/s^2 t = 2.00s v_0 = ? $y - y_0 = v_0 t + ½at^2$ $12.2-(36.3) = v_0(2) + ½(-9.8)(2)^2$ $-24.4 = v_0(2) - 19.6$ $v_0 = -2.4$ m/s	✓ ✓✓

(b) What is the velocity of the ball as it passes the top of the window?
Wat is die snelheid van die bal as dit by die bokant van die venster verby gaan?

$v = v_0 + at$ $= (-2.4) + (-9.8)(2)$ $= -22$ m/s Or $v^2 = v_0^2 + 2ay$ $= (-2.4)^2 + 2(-9.8)(12.2 - 36.6)$ $= 472.5$ $= \pm 22$ $= -22$ m/s	✓ ✓

Question 12 **[5]**

The figure shows the velocity time graph for a athlete. How far does the athlete travel in 16 s if v_s is 8.0 m/s?
Die figuur wys die snelheid-tyd grafiek vir 'n atleet. Hoe ver het die atleet beweeg in 16 s indien v_s is 8.0 m/s?



Distance = area under the curve t = 0 to 2 s: ½(2s)(8m) = 8 m ✓ t = 2 to 10 s: (8m)(8s) = 64 m ✓ t = 10 to 12 s: ½(2s)(4m) + (2s)(4m) = 12 m ✓ t = 12 to 16 s: (4m)(4s) = 16 m ✓ Total distance = 100 m ✓